

Unit 3, Lesson 3: Plotting, Ordering, and Comparing Rational Numbers – Part 1



Benchmark

MA.6.NSO.1.1 Extend previous understanding of numbers to define rational numbers. Plot, order, and compare rational numbers.



Learning Targets

- ☐ I can plot and compare rational numbers represented as a decimal or percent.
- ☐ I can plot, compare, and order fractional representations of rational numbers.



3.3.1 Warm-Up: Notice and Wonder

- The diagram gives the temperature, in Fahrenheit ($^{\circ}\text{F}$) and in Celsius ($^{\circ}\text{C}$), for Memphis, Tennessee, and Bangor, Maine.



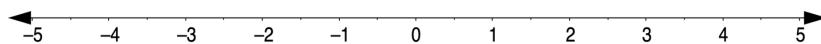
What do you notice about the temperatures?



3.3.2 Guided Instruction: Comparing and Ordering Decimals

1. The table shows the lowest temperature, in degrees Celsius ($^{\circ}\text{C}$), for each day during the week of October 14, 2019, in Anchorage, Alaska.

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Temperature ($^{\circ}\text{C}$)	4.2	-3.7	-0.75	1.3	2.8	-0.1	-2.6



- Plot the temperatures on the number line.
- Which temperature did you plot farthest to the left?
- What day of the week is represented by the value that was plotted farthest to the left?

- d. Why does the location of the temperature that is plotted farthest to the left help determine which day had the lowermost temperature?

- e. Interpret the temperature in this context.

During the week of October 14, 2019, the lowermost temperature was _____ $^{\circ}\text{C}$. This was the

- ☐ warmest
☐ coldest

- f. The inequality symbols, $<$ or $>$, can be used to compare two of the daily low temperatures. Write an inequality that shows the relationship between two of the low temperatures for the week of October 14, 2019.

2. Jayden, a data analyst, is comparing the populations of four different states to the population of the United States. His data is shown in the table.

State	Population as a Portion of the U.S. Population (decimal form)	Population as a Percentage of the U.S. Population (percent form)
California	0.119	
Florida		6.4%
New York		5.9%
Texas	0.087	

- a. Choose two states to compare. Write or describe your comparison.

- b. Olivia and Anthony's strategies for comparing California and New York are explained below.

Olivia explained, "I know that percent represents the part per 100, so the population of New York is approximately $\frac{6}{100}$ of the population of the U.S. The portion of the U.S. population living in California is approximately 12 hundredths, or $\frac{12}{100}$. Therefore, more people live in California than in New York."

Anthony explained, "I can convert 5.9% to a decimal by dividing 5.9 by 100. So 5.9% can also be represented by 0.059. I know 0.119 is greater than 0.059. Therefore, more people live in California than in New York."

Olivia and Anthony both used different methods to come to the same conclusion. Discuss their methods with a partner.

Olivia's Method	Anthony's Method

- c. Complete each inequality statement below to compare the given values.

0.119 ____ 6.4% 6.4% ____ 0.087 0.087 ____ 5.9%

- d. In which of the four states does the highest portion of the U.S. population live?

- e. In which of the four states does the smallest portion of the U.S. population live?



3.3.3 Exploration: Plotting and Comparing Fractions

1. Plot and label the rational numbers on the number line below.

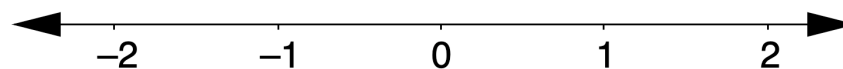
$$\frac{-3}{2}$$

$$\frac{5}{8}$$

$$\frac{5}{4}$$

$$\frac{1}{4}$$

$$\frac{-3}{4}$$



Using the rational numbers plotted above, answer the following questions.

2. Andre says that $\frac{1}{4}$ is less than $-\frac{3}{4}$ because out of the two values, $\frac{1}{4}$ is closer to 0. Explain why you agree or disagree with Andre.

3. Complete the table for each pair of numbers.

Pair	Farthest Distance From 0	Greater Value
$\frac{1}{4}$ or $\frac{5}{4}$		
$-\frac{3}{4}$ or $\frac{5}{8}$		

4. Is the value that is farther from 0 always the greater value? Explain your reasoning.



3.3.4 Your Turn!

1. Explain why each inequality statement is true or false.

a. $-5.2 < -7.1$

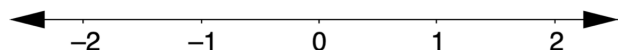
b. $3.1 > 38\%$

2. Use the symbols $<$, $>$ or $=$ to compare each pair of values.

a. $-\frac{4}{3}$ _____ $-\frac{3}{2}$ b. 0.75 _____ $\frac{3}{2}$ c. $\frac{3}{2}$ _____ 1.5

3. Plot the following values on the number line. Label each point with its numeric value.

$$1.5 \quad -\frac{3}{2} \quad -\frac{4}{3} \quad 0.75 \quad \frac{3}{2}$$



- a. Complete the inequality statement to show the relationship between the values.

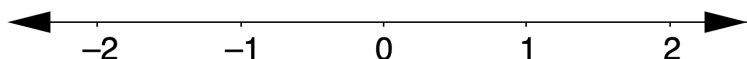
_____ > _____ > _____ > _____

4. Select all the values that are greater than -5 .

- ☐ 1.324
- ☐ -6.2
- ☐ -12.67
- ☐ 0.231
- ☐ -1.4
- ☐ -4.789

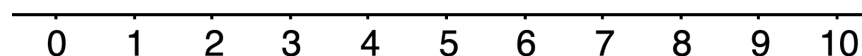
5. Order these values from least to greatest.

$$0.43, \quad -4\frac{1}{5}, \quad 0, \quad 50\%, \quad -\frac{3}{2}, \quad -6.712, \quad 19\%$$



3.3.5 Wrap-Up: Reflection

1. Circle the emoji that best represents your understanding of the lesson today.



2. Write an explanation of why you chose that emoji.